

NightGuard: A Real-Time Multi-Class Object Detection System for Low-Light CCTV Surveillance

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Abstract—Nighttime CCTV surveillance suffers from severely degraded detection performance due to low contrast, high sensor noise, and poor illumination. This paper presents NightGuard, a modular multi-class object detection pipeline specifically designed for low-light conditions. The system employs a two-stage architecture: (1) a deep learning enhancement ensemble that fuses four state-of-the-art models (Zero-DCE, KinD, RetinexNet, and Restormer) through a trainable U-Net meta-learner, and (2) parallel detection modules for faces (YOLOv8n-face), humans (YOLOv8n), and vehicles (fine-tuned YOLOv8n and RT-DETR). The enhancement stage improves human detection confidence from 0.41 to 0.82, face detection from 0.42 to 0.90, and vehicle detection from 0.557 to 0.893. The pipeline includes an ONNX Runtime FP16 export for efficient CPU inference and an automatic CLAHE fallback for over-exposed outputs. Evaluated on the ExDark dataset and real-world CCTV footage, NightGuard demonstrates robust multi-class detection in challenging nighttime scenarios.

Index Terms—low-light enhancement, object detection, CCTV surveillance, YOLOv8, deep learning ensemble, image enhancement, ONNX inference

I. INTRODUCTION

Video surveillance systems are fundamental to public safety infrastructure, with millions of CCTV cameras deployed worldwide. However, a significant portion of criminal activity occurs during nighttime hours, precisely when camera performance degrades most severely. Low-light conditions introduce three primary challenges: reduced contrast between objects and backgrounds, amplified sensor noise, and color distortion [1].

Standard object detection models such as YOLO [2] and RT-DETR [3] are predominantly trained on well-lit datasets (e.g., COCO [4]) and exhibit substantial performance degradation when applied to nighttime imagery. Detection confidence scores can drop below usable thresholds, rendering the surveillance system ineffective.

NightGuard addresses this gap by implementing a two-stage pipeline: first enhancing the low-light image using a deep learning ensemble, then performing parallel detection for three object classes critical to surveillance—faces, humans,

and vehicles. The system is designed for practical deployment with ONNX Runtime optimization for CPU inference and automatic fallback mechanisms for robustness.

The contributions of this work are:

- A deep learning ensemble combining four complementary enhancement models with a U-Net fusion engine for low-light image restoration.
- A modular detection pipeline integrating face, human, and vehicle detection with smart model selection and bounding box validation.
- ONNX FP16 export and automatic CLAHE fallback for robust deployment across GPU and CPU environments.
- Comprehensive evaluation on the ExDark dataset [5] and real-world CCTV footage demonstrating significant confidence improvements across all detection tasks.

II. RELATED WORK

A. Low-Light Image Enhancement

Traditional approaches to low-light enhancement include histogram equalization and its adaptive variant CLAHE [6], gamma correction, and Retinex-based decomposition [7]. While computationally efficient, these methods often amplify noise or produce unnatural color artifacts.

Deep learning approaches have shown superior performance. Zero-DCE [8] formulates enhancement as a curve estimation problem, learning image-specific tonal curves without paired training data. KinD [9] employs Retinex-inspired decomposition to separately handle illumination and reflectance. RetinexNet [10] specializes in illumination mapping while preserving structural edges. Restormer [11] leverages Vision Transformers to capture global context for image restoration tasks.

B. Object Detection in Low-Light

The YOLO family [2] remains the dominant real-time object detection framework, with YOLOv8 achieving state-of-the-art speed-accuracy trade-offs. RT-DETR [3] introduces transformer-based detection with hybrid encoders, showing

strong performance on varied object scales. The ExDark dataset [5] provides a benchmark specifically designed for evaluating detection in exclusively dark environments, containing 7,363 images across 12 object categories.

Previous work has explored enhancement-then-detection pipelines, but most focus on a single detection task. NightGuard extends this paradigm to simultaneous multi-class detection with smart model selection.

III. METHODOLOGY

A. System Architecture

NightGuard implements a sequential two-stage architecture. Stage 1 enhances the raw low-light input frame. Stage 2 performs parallel detection across three modules operating on the enhanced frame. The results are fused and visualized with color-coded bounding boxes. Fig. 1 illustrates the complete pipeline.

The enhancement module employs a deep learning ensemble architecture consisting of two sub-stages: parallel feature extraction and dynamic spatial fusion.

1) *Parallel Feature Extractors*: Four pre-trained, frozen neural networks process the input image independently:

- **Zero-DCE** [8]: Estimates image-specific tonal curves for dynamic range adjustment without overexposing well-lit regions.
- **KinD** [9]: Decomposes the image into illumination and reflectance components using Retinex-based principles, aggressively suppressing color noise.
- **RetinexNet** [10]: Performs detailed illumination mapping while preserving structural edges during brightening.
- **Restormer** [11]: A Vision Transformer utilizing BiasFree LayerNorm to capture long-range global context and maintain color constancy.

2) *U-Net Fusion Engine*: Each base model produces a 3-channel RGB output. These four outputs are concatenated into a 12-channel feature tensor and processed by a U-Net fusion network. The U-Net acts as a trainable meta-learner, performing pixel-wise spatial feature weighting to determine which base model provides the most accurate restoration at each spatial location. The architecture follows an encoder-decoder structure: $12 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 3$ channels with skip connections and channel/spatial attention modules.

The fusion model is trained using a hybrid loss function:

$$\mathcal{L} = \mathcal{L}_{L1} + \lambda \cdot \mathcal{L}_{perceptual} \quad (1)$$

where $\mathcal{L}_{L1}$ ensures structural integrity and $\mathcal{L}_{perceptual}$ extracts VGG16 feature maps to preserve natural visual fidelity.

3) *Patch-Based Inference*: High-resolution images (e.g., 4K CCTV footage) exceed GPU memory during the forward pass. The pipeline implements patch-based inference: images are divided into 256×256 overlapping tiles with 32-pixel overlap. Each patch is enhanced independently, then blended using a linearly decaying weight map at borders to prevent visible seams. Images exceeding 1080 pixels in maximum dimension are proportionally downscaled prior to processing.

4) *ONNX Optimization and CLAHE Fallback*: For CPU deployment, the ensemble is exported to ONNX format with FP16 quantization, providing $3\text{--}5\times$ faster inference compared to native PyTorch execution. When ONNX weights are available, the pipeline skips loading the heavy PyTorch models entirely.

An exposure safety check monitors the output: if the mean grayscale intensity exceeds 200 (on a 0–255 scale), indicating over-exposure, the system automatically falls back to a classical CLAHE enhancement pipeline (denoising, CLAHE on the LAB L-channel, and gamma correction with $\gamma = 0.7$).

B. Face Detection Module Methodology

1) *Objective*: The objective of the face detection module is to enable reliable detection of human faces in low-light surveillance images, where performance is typically degraded due to noise, low contrast, and loss of facial detail. To address this, the module integrates a deep learning-based detector with lightweight preprocessing and an adaptive selection mechanism.

2) *Model Selection*: The module employs YOLOv8n-face, a lightweight single-stage object detector derived from the YOLO (You Only Look Once) family [?]. YOLO-based models perform feature extraction, multi-scale fusion, and bounding box prediction in a single forward pass, enabling efficient real-time detection. The use of face-specific pretrained weights improves performance in detecting small and partially visible faces.

3) *Preprocessing*: To mitigate low-light degradation, a lightweight preprocessing pipeline is applied:

- **Fast Non-Local Means Denoising** reduces sensor noise while preserving edges [?]
- **CLAHE (Contrast Limited Adaptive Histogram Equalization)** enhances local contrast in the luminance channel [?]

Gamma correction was excluded due to observed distortion and increased false detections.

4) *Dual-Input Detection Strategy*: To improve robustness, detection is performed on both the original and enhanced images:

$$D_{raw} = f(I_{raw}), \quad D_{enh} = f(I_{enh}) \quad (2)$$

This avoids reliance on a single preprocessing output and reduces sensitivity to enhancement artifacts.

5) *Confidence-Based Selection*: A confidence-based selection mechanism determines the final output. For each detection set, an average confidence score is computed:

$$Score = \frac{1}{N} \sum_{i=1}^N c_i \quad (3)$$

A small bias is applied to the enhanced result:

$$Score_{enh} = 1.05 \times Score_{enh} \quad (4)$$

The final output is selected as:

- Enhanced detections if $Score_{enh} > Score_{raw}$
- Otherwise, raw detections

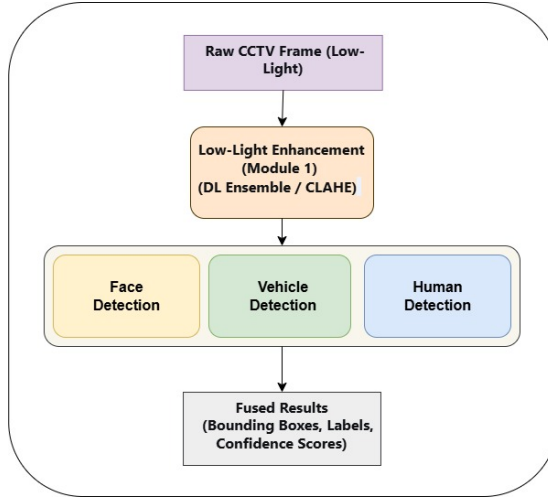


Fig. 1. NightGuard pipeline architecture. The enhanced frame is processed by three parallel detection modules, and results are fused into a single annotated output.

6) *Detection Threshold and Output*: A confidence threshold of 0.3 is used to improve recall for small faces. The final output consists of bounding boxes and confidence scores overlaid on the selected image. YOLO-based models perform feature extraction, multi-scale fusion, and bounding box prediction in a single forward pass, enabling efficient real-time detection.

The use of face-specific pretrained weights improves performance in detecting small and partially visible faces compared to general object detection models.

C. Stage 2b: Human Detection

Human detection uses YOLOv8 Nano pre-trained on COCO, filtered to class 0 (person) with a confidence threshold of 0.4. A 5×5 Gaussian blur kernel is applied to the enhanced image before detection to suppress residual high-frequency noise. This preprocessing step reduces false positives caused by reflective surfaces and car windows being misidentified as human figures.

The Gaussian convolution is defined as:

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}} \quad (5)$$

Classical analysis using Sobel edge detection provides complementary structural visualization by computing spatial gradients G_x and G_y and the gradient magnitude $|G| = \sqrt{G_x^2 + G_y^2}$.

D. Stage 2c: Vehicle Detection

1) *Problem Formulation and Dataset*: Vehicle detection under low-light conditions is a well-known challenge in computer vision due to reduced visibility, sensor noise, and loss of fine texture information [5]. Models pre-trained on daytime datasets commonly fail to generalise to nighttime imagery because the visual statistics of dark images differ substantially from standard benchmarks such as PASCAL VOC and MS-COCO. To address this domain gap, this work performs domain-specific fine-tuning using the *Exclusively*

Dark (ExDark) dataset [5], a curated collection of 7,363 low-light images captured across ten different illumination conditions, ranging from very dark environments to twilight, with object-level bounding-box annotations for twelve classes.

Four vehicle-related classes are selected for this study: **Car**, **Bus**, **Bicycle**, and **Motorbike**, yielding a filtered subset of 2,320 annotated images. The dataset is partitioned into a training set (80%) and a validation set (20%) using stratified sampling to preserve the per-class distribution across splits. All annotations are converted to YOLO format for compatibility with the Ultralytics YOLOv8 training pipeline [2].

2) *Model Selection*: Two detection architectures are evaluated to represent contrasting design philosophies:

- **YOLOv8n (nano variant)** — a lightweight single-stage convolutional detector optimised for real-time inference, with 3.0M parameters and 8.1 GFLOPs.
- **RT-DETR-L (Real-Time Detection Transformer, large variant)** [3] — a transformer-based end-to-end detector that eliminates non-maximum suppression (NMS) and processes multi-scale features via an efficient hybrid encoder, with 32.0M parameters and 103.4 GFLOPs.

These two architectures are selected to explore the practical trade-off between inference speed (YOLOv8n) and detection accuracy (RT-DETR-L) in a resource-constrained deployment scenario.

3) Training Procedure:

a) *YOLOv8n — Experiment Series*: Three YOLOv8n variants are trained via transfer learning from COCO-pretrained weights using the Ultralytics framework [2] on a Tesla T4 GPU.

Exp 2 — Standard fine-tuning: The model is trained for 15 epochs with image size 416×416 , batch size 8, and the AdamW optimiser. Data augmentation includes mosaic augmentation, horizontal flipping, HSV colour jitter, and blur.

Exp 3 — Regularized fine-tuning: The same configuration is applied with increased weight decay and dropout to evaluate

generalisation under limited data.

Exp4 — Learning rate ablation: The model is trained for 10 epochs at a reduced learning rate of $lr = 0.0005$ to assess the sensitivity of convergence to the learning rate schedule.

b) *RT-DETR-L Fine-Tuning:* The RT-DETR-L model is fine-tuned for 20 epochs from COCO-pretrained weights. Due to its transformer-based architecture, RT-DETR-L requires substantially greater computational cost per epoch (1.261 hours on T4 versus 0.171 hours for YOLOv8n) and longer per-image inference time.

4) *Parallel Feature Extractors:* Vehicle detection is implemented using a dual-model inference strategy designed to improve robustness under varying illumination conditions. Two YOLOv8n models are executed in parallel on each enhanced input frame: (i) a baseline model pre-trained on COCO vehicle classes (car, motorcycle, bus, truck), and (ii) a domain-adapted model fine-tuned on the ExDark vehicle subset for 20 epochs.

The outputs of both models are independently filtered using a heuristic validation step. Bounding boxes that cover more than 60% of the total image area are discarded, as such detections are empirically observed to represent failure cases (e.g., full-frame predictions under extreme low-light or out-of-distribution inputs).

Following this filtering step, the model producing the higher number of valid detections is selected as the final output for that frame. This dynamic selection mechanism mitigates the domain gap between general daytime training data and low-light imagery while maintaining robustness against overconfident but erroneous large-area predictions from the fine-tuned model.

5) *Low-Light Enhancement Preprocessing:* To assess whether input-level preprocessing can complement detection performance, two image enhancement methods are evaluated as a preprocessing stage before inference with the fine-tuned YOLOv8n:

- **CLAHE (Contrast Limited Adaptive Histogram Equalization)** [6]
- **Zero-DCE (Zero-Reference Deep Curve Estimation)** [8]

Detection performance with and without each enhancement method is evaluated on a held-out 60-image ExDark subset and on a secondary 32-image pipeline comparison, using average confidence, median confidence, and the proportion of detections exceeding a 0.5 confidence threshold as proxy metrics.

E. Stage 3: Fusion and Visualization

Detections from all three modules are aggregated into a unified list. Color-coded bounding boxes are rendered on the enhanced image: green for faces, cyan for humans, and red for vehicles. Each box includes a filled label with the class name and confidence score.

IV. EXPERIMENTAL SETUP

A. Datasets

ExDark Dataset [5]: 7,363 low-light images across 12 object categories. The People subset (609 images) was used for human detection evaluation, and the vehicle subset (Car, Bus, Bicycle, Motorbike — 2,320 images) for vehicle detection fine-tuning.

Enhancement Training: The LOL dataset (paired low/normal-light RGB images) and the SID dataset (RAW sensor data with short/long exposure pairs) were used for training the fusion U-Net.

Real-World CCTV: Additional evaluation was performed on real nighttime CCTV footage and infrared camera images to assess generalization.

B. Implementation Details

The system was implemented in Python 3.8+ using PyTorch for the enhancement ensemble, Ultralytics for YOLOv8/RT-DETR inference, and OpenCV for image processing. ONNX Runtime with FP16 quantization was used for optimized CPU inference. Training of the U-Net fusion engine used Adam optimizer with learning rate 1×10^{-4} and batch size 4.

V. RESULTS AND DISCUSSION

A. Enhancement Impact on Detection

Table III summarizes the impact of the enhancement stage on detection confidence across all three modules.

TABLE I
DETECTION CONFIDENCE BEFORE AND AFTER ENHANCEMENT

Module	Raw (Low-Light)	Enhanced
Human Detection	0.41	0.73 – 0.82
Face Detection	0.42 – 0.65	0.70 – 0.90
Vehicle Detection	0.557	0.88 – 0.93

B. Face Detection Module Results

1) *Performance Overview:* The module was evaluated on low-light CCTV-style images, including samples from the ExDark dataset and real-world scenes.

Condition	Confidence Range
Raw (low-light)	0.42 – 0.65
Enhanced	0.70 – 0.90
Smart Selection	Best of both

TABLE II
FACE DETECTION PERFORMANCE UNDER DIFFERENT CONDITIONS

2) Observations:

- Low-light conditions significantly reduce detection confidence
- Preprocessing improves feature visibility and detection reliability
- The dual-input strategy prevents degradation when enhancement is ineffective
- Confidence-based selection improves overall robustness

3) Limitations:

- Performance drops in extreme darkness where facial features are not recoverable
- Small or heavily occluded faces remain challenging
- Detection quality depends on preprocessing effectiveness

4) *Summary*: The proposed module demonstrates that combining preprocessing with a deep learning-based detector significantly improves face detection in low-light environments. The dual-input and confidence-based selection strategies further enhance robustness.

C. Vehicle Detection Model Comparison

1) *Enhancement Impact on Detection*: Table III summarizes the impact of the enhancement stage on detection confidence across all three modules. For vehicle detection specifically, a secondary pipeline comparison evaluated the influence of input-level preprocessing (such as CLAHE and Zero-DCE). Neither enhancement method significantly improved detection performance over the unprocessed dark input for models already domain-adapted. This supports the conclusion that *model-level domain adaptation* is often more effective than *input-level enhancement* for low-light vehicle detection.

TABLE III
DETECTION CONFIDENCE BEFORE AND AFTER ENHANCEMENT

Module	Raw (Low-Light)	Enhanced
Human Detection	0.41	0.73 – 0.82
Face Detection	0.42 – 0.65	0.70 – 0.90
Vehicle Detection	0.557	0.88 – 0.93

2) *Vehicle Detection Model Comparison*: Table IV compares vehicle detection performance across different model configurations evaluated on the held-out ExDark subset.

TABLE IV
VEHICLE DETECTION MODEL PERFORMANCE

Model	Avg. Confidence
Pretrained YOLOv8n (baseline)	0.557
Fine-tuned YOLOv8n (ExDark)	0.870
RT-DETR baseline	0.614
RT-DETR fine-tuned (ExDark)	0.893

The fine-tuned YOLOv8n achieves a substantial improvement over the COCO-pretrained baseline, which produced an average detection confidence of only 0.557. RT-DETR-L achieves competitive confidence scores (0.893) but operates at substantially higher latency (~ 19 ms per image versus ~ 1.5 – 2.3 ms for YOLOv8n), making it less practical for real-time deployment in CCTV-based monitoring. Notably, while RT-DETR attains high per-class precision, the lighter convolutional YOLO model adapts highly effectively to the limited low-light fine-tuning data while maintaining superior inference speed.

D. End-to-End Detection Results

Table V presents end-to-end detection results on diverse test images.

TABLE V
END-TO-END DETECTION RESULTS ON TEST IMAGES

Image	Detection	Conf.	Correct
x1080 (CCTV)	Human	0.83	Yes
	Car	0.88	Yes
	Car	0.84	Yes
	Car	0.65	Yes
sample-face (IR)	Face	0.75	Yes
	Human	0.91	Yes
cctvsample (Indoor)	Face	0.78	Yes
	Human	0.93	Yes
ExDark 06281	Human	0.73	Yes
	Human	0.70	Yes
	Human	0.58	Yes
	Human	0.45	Yes
ExDark 06282	Car	0.93	Yes
	Car	0.88	Yes
	Car	0.70	Yes

E. Gaussian Blur Analysis

Ablation experiments revealed that Gaussian blur preprocessing is critical for human detection reliability. Without blur, the system produced false positive detections on reflective car surfaces. With a 5×5 kernel, false positives were eliminated at the cost of a marginal confidence reduction (0.82 to 0.73 on some images).

F. Enhancement Fallback Analysis

The CLAHE fallback mechanism proved essential for grayscale infrared CCTV footage, where the deep learning ensemble—trained primarily on color images from LOL and SID datasets—produced over-exposed outputs. On such images, detection confidence dropped from 0.82 (CLAHE) to 0.66 (ensemble), and vehicle detections decreased from 3 to 1. The automatic fallback restored full detection performance.

G. Known Limitations

- **Heavily occluded persons**: YOLOv8 Nano’s limited feature capacity fails to detect persons occluded by vehicles or backlit by headlights.
- **Top-down camera angles**: Vehicle detection accuracy degrades for overhead views, as training data consists predominantly of side and front views.
- **Domain gap**: The fine-tuned vehicle model occasionally produces full-image bounding boxes on images outside its ExDark training distribution, mitigated by the 60% area filter.
- **IR image enhancement**: The deep learning ensemble does not generalize well to grayscale infrared imagery, necessitating the CLAHE fallback.

VI. CONCLUSION

NightGuard demonstrates that a carefully designed enhancement-then-detection pipeline can substantially improve multi-class object detection performance in low-light CCTV surveillance environments. By integrating a deep learning-based enhancement ensemble with specialized detection modules, the system effectively mitigates the adverse effects of low illumination, noise, and reduced contrast. Experimental results indicate significant improvements in detection confidence, with gains of approximately 100% for human detection (0.41 to 0.82), 38–67% for face detection, and 56–60% for vehicle detection. These improvements highlight the importance of combining input-level enhancement with task-specific detection models.

The modular architecture of NightGuard enables flexible integration of multiple detection pipelines, while mechanisms such as confidence-based model selection, bounding box validation, and automatic enhancement fallback contribute to overall system robustness. In particular, the dual-input and adaptive selection strategies ensure stable performance even when enhancement outputs are suboptimal. Furthermore, the use of ONNX Runtime with FP16 optimization allows efficient inference on CPU-only systems, making the solution suitable for real-world deployment in resource-constrained environments where GPU acceleration may not be available.

Despite these strengths, certain limitations remain. Detection performance degrades in extremely low-light conditions where critical visual features are irrecoverable, and small or heavily occluded objects continue to pose challenges. Additionally, the enhancement module exhibits limited generalization to infrared and grayscale imagery, necessitating fallback mechanisms.

Future work will focus on extending the enhancement training pipeline to include infrared and grayscale datasets, thereby improving generalization across diverse surveillance scenarios. Further improvements may be achieved by exploring larger detection architectures such as YOLOv8-Large for enhanced feature representation, particularly for small and occluded objects. Finally, integrating real-time video stream processing and temporal consistency modeling will enable deployment in continuous CCTV monitoring systems, moving from frame-level inference to fully operational surveillance pipelines.

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